

# Homework 1

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## Problem 1

### Problem Statement

Let  $a$  and  $b$  be integers, Prove that if  $2 + a$  and  $35 - b$  are divisible by 11, then  $a + b$  is divisible by 11.

### Solution

*Proof.* Since  $2 + a$  and  $35 - b$  are divisible by 11, it can be represented with the following equations:

$$\begin{aligned}2 + a &= 11 \cdot p, & p &\in \mathbb{Z} \\35 - b &= 11 \cdot q, & q &\in \mathbb{Z}\end{aligned}$$

where  $p$  and  $q$  are integers.

Each equations can be rearranged as follows.

$$\begin{aligned}a &= 11 \cdot p - 2 \\b &= 35 - 11 \cdot q\end{aligned}$$

The sum  $a + b$  is then as follows.

$$\begin{aligned}a + b &= 11p - 2 + 35 - 11q \\&= 11 \cdot (p - q) + 33 \\&= 11 \cdot (p - q) + 11 \cdot 3 \\&= 11 \cdot (p - q + 3) \\&= 11 \cdot k & k = p - q + 3 \in \mathbb{Z}\end{aligned}$$

$k$  is guaranteed to be an integer since it is a sum of integers. Since  $a + b$  is a product of 11 and some integer  $k$ ,  $a + b$  is divisible by 11.

□

## Problem 2

Textbook p15 (15.c, 15.e)

### Problem Statement

Use Euclidean algorithm to find:

(a)  $(1003, 456)$

(b)  $(5858, 1436)$

### Solution

(a)

$$\begin{aligned}(1003, 456) &= (456, 91) \\ &= (91, 1) \\ &= (1, 0) \\ &= 1\end{aligned}$$

$$\begin{aligned}1003 &= 456 \times 2 + 91 \\ 456 &= 91 \times 5 + 1 \\ 91 &= 1 \times 91 + 0\end{aligned}$$

Answer:  $(1003, 456) = 1$

(b)

$$\begin{aligned}(5858, 1436) &= (1436, 114) \\ &= (114, 68) \\ &= (68, 46) \\ &= (46, 22) \\ &= (22, 2) \\ &= (2, 0) \\ &= 2\end{aligned}$$

$$\begin{aligned}5858 &= 1436 \times 4 + 114 \\ 1436 &= 114 \times 12 + 68 \\ 114 &= 68 \times 1 + 46 \\ 68 &= 46 \times 1 + 22 \\ 46 &= 22 \times 2 + 2 \\ 22 &= 2 \times 11 + 0\end{aligned}$$

Answer:  $(5858, 1436) = 2$

## Problem 3

Textbook p15 (15.j)

### Problem Statement

Express  $(1003, 456)$  as a linear combination of 1003 and 456.

## Solution

From using the Euclidean algorithm in Problem 2 (a), following equations are found.

$$1003 = 456 \times 2 + 91 \quad (1)$$

$$456 = 91 \times 5 + 1 \quad (2)$$

These equations can be rearranged as follows.

$$91 = 1003 - 456 \times 2 \quad (3)$$

$$1 = 456 - 91 \times 5 \quad (4)$$

By substituting Equation (3) to Equation (4), 1 can be represented as a linear combination of 1003 and 456.

$$\begin{aligned} 1 &= 456 - (1003 - 456 \times 2) \times 5 \\ &= 456 \times 11 - 1003 \times 5 \end{aligned} \quad (5)$$

The GCD of 1003 and 456 is 1, which can be represented as a linear combination of 1003 and 456 as follows.

$$1 = 456 \times 11 - 1003 \times 5$$

## Problem 4

### Problem Statement

Show that for any positive integer  $n$ , the fraction  $\frac{12n+1}{30n+2}$  cannot be reduced.

### Solution

*Proof.* For a fraction to be non-reducible, the denominator and the numerator must be coprime. In other words, the GCD of the denominator and the numerator must be 1.

The problem may be reformulated as follows: show that for any positive integer  $n$ ,  $(12n+1), 30n+2$  is 1. Note that since  $n$  is a positive integer,  $12n+1$  is always less than  $30n+2$ .

Applying the Euclidean algorithm results in the following:

$$\begin{aligned} (30n+2, 12n+1) &= (12n+1, 6n) \\ &= (6n, 1) \\ &= (1, 0) \\ &= 1 \end{aligned}$$

From the above result, it is shown that the GCD of  $30n + 2$  and  $12n + 1$  is 1, meaning that  $30n + 2$  and  $12n + 1$  are coprime. Therefore, the fraction  $\frac{12n+1}{30n+2}$  cannot be reduced for any positive integer  $n$

□

## Problem 5

### Problem Statement

Find  $(2^{100} - 1, 2^{120} - 1)$ .

### Solution

Let  $q \in \mathbb{Z}^+$  be the GCD of  $2^{100} - 1$  and  $2^{120} - 1$ . This can be represented as the following.

$$\begin{aligned}(2^{100} - 1) \bmod q &\equiv 0 \bmod q \\ (2^{120} - 1) \bmod q &\equiv 0 \bmod q\end{aligned}$$

By modulus arithmetic, the relations can be rearranged as following.

$$\begin{aligned}2^{100} \bmod q &\equiv 1 \bmod q \\ 2^{120} \bmod q &\equiv 1 \bmod q\end{aligned}$$

Then, the two relations can be combined.

$$2^{100} \bmod q \equiv 2^{120} \bmod q$$

Divide both sides by  $2^{100} \bmod q$  and rearrange.

$$\begin{aligned}2^{20} \bmod q &\equiv 1 \bmod q \\ (2^{20} - 1) \bmod q &\equiv 0 \bmod q\end{aligned}$$

$q$  must divide  $2^{20} - 1$ . The largest number that can achieve this is  $2^{20} - 1$ . Therefore, the GCD of  $(2^{100} - 1, 2^{120} - 1)$  is  $2^{20} - 1$ .

## Problem 6

Textbook p16 (29)

## Problem Statement

Prove that a positive integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

## Solution

First, we prove that if a positive integer is divisible by 9, its sum of digits is divisible by 9.

*Proof.* Let  $d_0, d_1, d_2, \dots, d_{n-1}$  be the digits of a  $n$  digit positive number  $k$  where  $k = d_0 + 10^1 \cdot d_1 + 10^2 \cdot d_2 + \dots + 10^{n-1} \cdot d_{n-1}$ . We would like to prove that if 9 divides  $k$ , 9 divides sum of  $d_0, d_1, d_2, \dots, d_{n-1}$ .

$k$  can be reformulated as follows:

$$\begin{aligned} k &= d_0 + 10^1 \cdot d_1 + 10^2 \cdot d_2 + \dots + 10^{n-1} \cdot d_{n-1} \\ &= \sum_{i=0}^{n-1} 10^i \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + (10^i - 1) \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + \sum_{i=0}^{n-1} (10^i - 1) \cdot d_i \end{aligned}$$

By factoring  $10^i - 1$ ,  $k$  can be further reformulated as follows:

$$\begin{aligned} k &= \sum_{i=0}^{n-1} d_i + \sum_{i=0}^{n-1} (10^i - 1) \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + \sum_{i=0}^{n-1} (10 - 1)(10^{i-1} + 10^{i-2} + \dots + 10^1 + 1) \cdot d_i \\ &= \sum_{i=0}^{n-1} d_i + 9 \sum_{i=0}^{n-1} (10^{i-1} + 10^{i-2} + \dots + 10^1 + 1) \cdot d_i \end{aligned}$$

Now  $k$  is in the form  $k = ab + r$ , where  $r = \sum_{i=0}^{n-1} d_i$ ,  $a = 9$ , and  $b = \sum_{i=0}^{n-1} (10^{i-1} + 10^{i-2} + \dots + 10^1 + 1) \cdot d_i$ . Since 9 divides  $k$ , 9 also must divide  $r$ , which is the sum of digits. Therefore, if a positive integer is divisible by 9, sum of its digits are also divisible by 9.  $\square$

Next, we prove that if the sum of a positive integer's digits is divisible by 9, the positive integer is also divisible by 9.

*Proof.* From previous formulation of  $k$ , it is established that  $k = ab + r$  where  $a = 9$ . Since  $r$  is the sum of digits and can be represented in the form  $r = 9p$  for some integer  $p$ ,  $k$  can be reformulated as follows:

$$\begin{aligned} k &= ab + r \\ &= 9b + 9p \\ &= 9(b + p) \end{aligned}$$

This shows that  $k$  is a multiple of 9, meaning that it is divisible by 9.  $\square$

## Problem 7

Textbook p16 (31.b)

### Problem Statement

If  $t$  is an integer such that  $a_i \mid t$  for  $i = 1, 2, \dots, k$ , prove that  $[a_1, a_2, \dots, a_k] \mid t$ .

### Solution

*Proof.* Let  $m = [a_1, a_2, \dots, a_k]$ . By division algorithm,  $t = mq + r$ . For  $t$  to be divisible by  $m$ ,  $r$  must be 0.

For every  $i$ , let  $t = a_i u_i$  and  $m = a_i v_i$  since  $t$  and  $m$  are divisible by  $a_i$ . By rearranging the equation found by division algorithm,  $r = t - mq$ . Substituting  $t$  and  $m$  to this equation results in the following:  $r = a_i(u_i - v_i q)$ . Therefore,  $r$  is divisible by  $a_i$  for all  $i$   $\square$

## Problem 8

Textbook p23 (21)

### Problem Statement

If  $c^2 = ab$  and  $(a, b) = 1$ , where  $a, b, c$  are positive integers, prove that  $a$  and  $b$  are perfect squares.

### Solution

*Proof.* Assume contradiction: if  $c^2 = ab$  and  $(a, b) = 1$ , then  $a$  and  $b$  are not perfect squares.

From the second equation, it can be deduced that  $a$  and  $b$  are coprime, meaning that  $a$  does not equal to  $b$ .

Square rooting the first equation yields the following equation:  $c = \sqrt{ab} = \sqrt{a}\sqrt{b}$ . Since  $a$  does not equal to  $b$ ,  $a$  and  $b$  multiplied cannot result in a perfect

square. However, since  $a$  and  $b$  are positive integers, square root of them is irrational. This is a contradiction, as  $c$  must also be a positive integer.

Therefore, by contradiction, if  $c^2 = ab$  and  $(a, b) = 1$ ,  $a$  and  $b$  are perfect squares.  $\square$

## **Problem 9**

**Problem Statement**

**Solution**

## **Problem 10**

**Problem Statement**

**Solution**

## **Problem 11**

**Problem Statement**

**Solution**

## **Problem 12**

**Problem Statement**

**Solution**